

NI TB Genomic Surveillance

Outbreak Investigation Report

Generated: 2026-05-09 12:01:53 UTC

Case Summary

Total Cases	24
Clustered Cases	8
Unclustered Cases	16
Open Clusters	3

Automated Interpretation Flags

- Sequencing coverage is below target: 50.0% (target >= 80%).
- Open clusters requiring investigation: 3.
- High-confidence transmission links present; prioritize epidemiology follow-up.

Analysis Summary

Posterior Samples	1500
MCMC Iterations	2000
Burn-in	500
Mean Log-Likelihood	-25346.24
Log-Likelihood SD	5506.38

Programme Surveillance KPIs (Last 12 Weeks)

Eligible Cases	2
Sequenced Cases	1
Sequencing Coverage (%)	50.0
QC Reported Cases	0
QC Pass Cases	0
QC Fail Cases	0
QC Pass Rate (%)	None
Contamination Flags	0
Median Days Specimen to QC	None

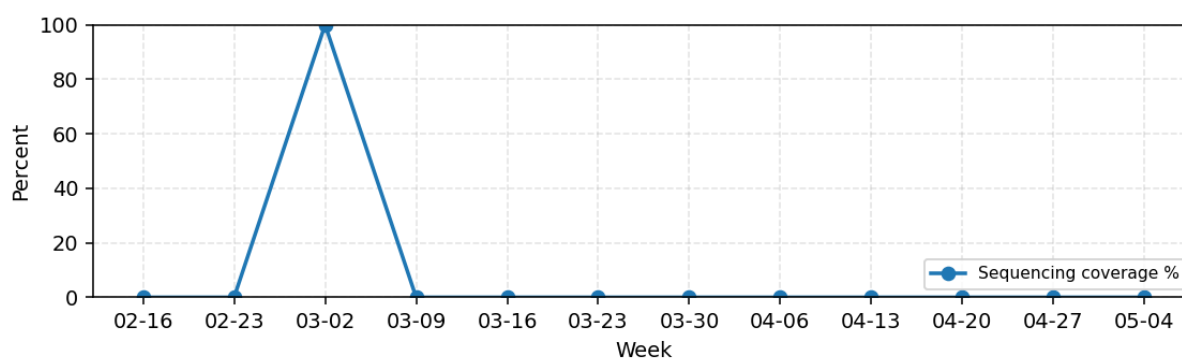
Regional Sequencing Representativeness

Region	Eligible	Sequenced	Coverage %
United Kingdom	2	1	50.0

Weekly Surveillance Trends (12 Weeks)

Week	Eligible	Sequenced	Coverage %	QC Pass %
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2026-02-16	0	0	None	None
2026-02-23	0	0	None	None
2026-03-02	1	1	100.00	None
2026-03-09	0	0	None	None
2026-03-16	0	0	None	None
2026-03-23	0	0	None	None
2026-03-30	0	0	None	None
2026-04-06	0	0	None	None
2026-04-13	0	0	None	None
2026-04-20	0	0	None	None
2026-04-27	0	0	None	None
2026-05-04	1	0	0.00	None



Cluster Action Prioritization

Cluster	Cases	Regions	Most Recent	Status	Priority
ee385d62	4	1	2025-08-02	open	14
8b3d4763	2	1	2025-03-14	open	10
1d21d293	2	1	2025-02-26	open	10

Priority score combines cluster size, cross-region spread, specimen recency, and open investigation status.

Lineage and Drug Resistance Validation

Overall Status	blocked_tool_dependencies
TB-Profiler	installed_but_unusable
Mykrobe	not_installed
Docker Fallback	True
Docker Daemon Running	False
FASTA Inputs	1

Lineage/DR Next Steps

- Install tb-profiler/mykrobe dependencies (for Windows this is often easiest in WSL2 or Conda).
- If local dependencies fail, install Docker Desktop and rerun with TBPROFILER_DOCKER_FALLBACK=1.
- Place FASTQ/VCF/FASTA inputs in uploads/ (or export pipeline outputs there).
- Optionally provide exports/lineage_resistance_calls.csv to hydrate tb_interpretation.

Secondary Transmission Engines

Secondary Validation Status	completed
Consensus State	pending_secondary_outputs
TransPhylo	not_installed
BactDating	not_installed
Tree Input Available	False
Cases Input Available	True

Cross-Method Clustering Comparison

Sequence Assigned Cases	8
Outbreaker Assigned Cases	24
Overlap Cases	8
Pairwise Precision	0.286
Pairwise Recall	1.0
Pairwise Jaccard	0.286

Sequence Clustering Snapshot

Cluster Count	3
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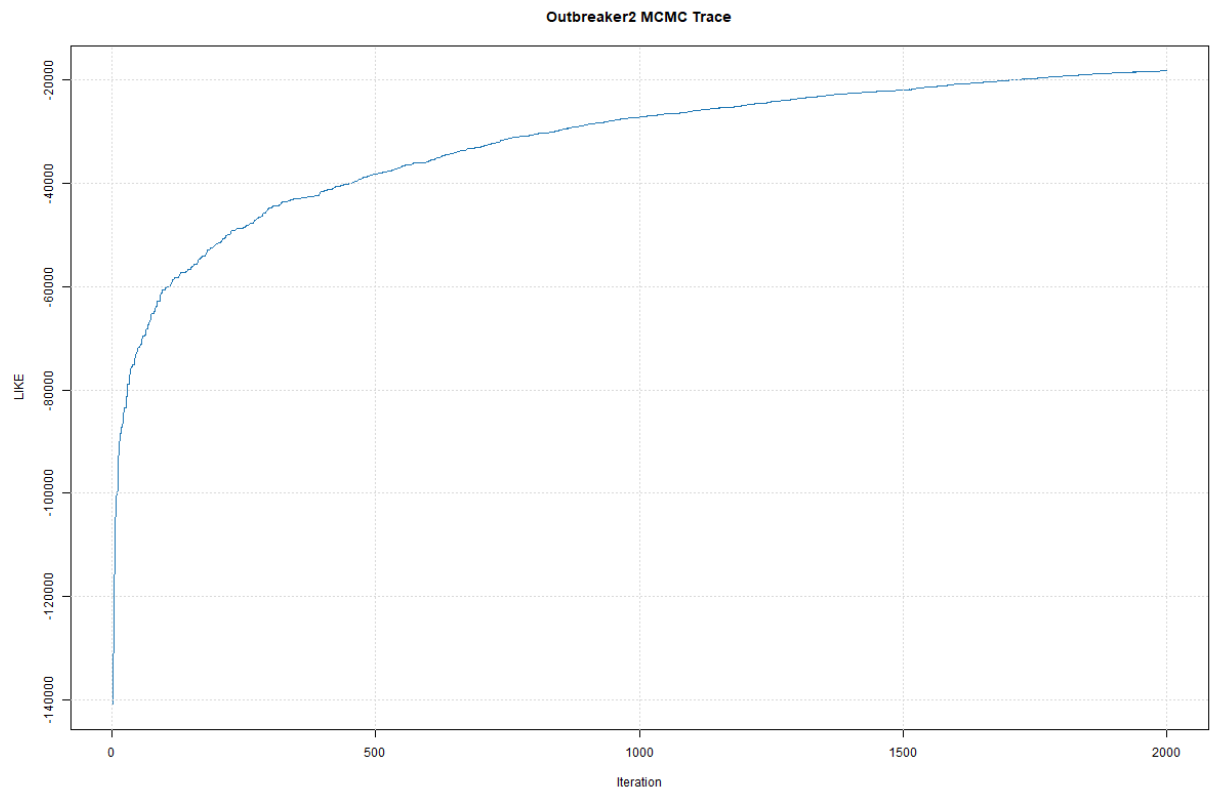
Transmission Priority Signals

Case	Region	Risk	Out	In
08eea384		3.6936	5	1
87ef58ef		2.7874	4	0
de93a139		2.3137	3	1
33eee0f9		1.7	2	1
66c038c2		1.6837	2	1
8a3f5b4d		1.6011	2	1
5c2b5eba		1	1	1
aed8a068		1	1	1
d154e9df		1	1	1
e29ce654		0.9946	1	1

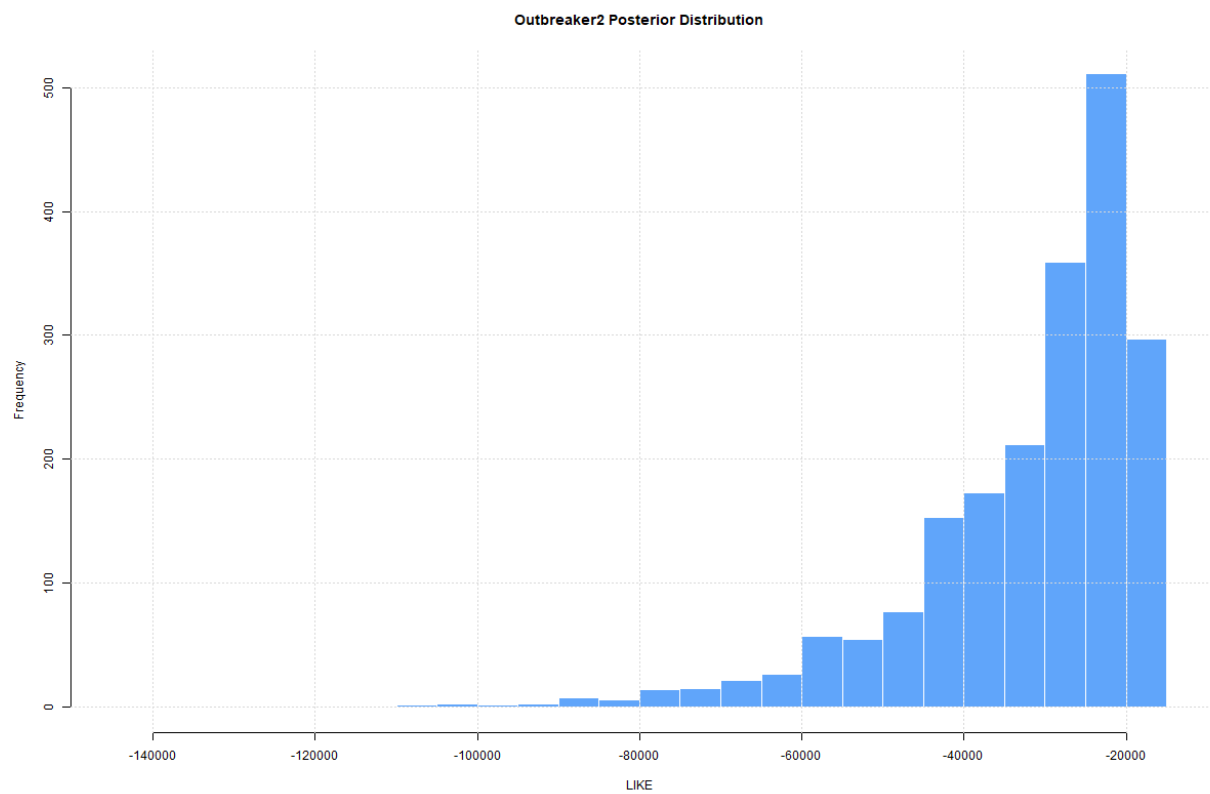
Network snapshot: nodes=24, edges=23, high-confidence links=22

Diagnostic Graphics

Trace

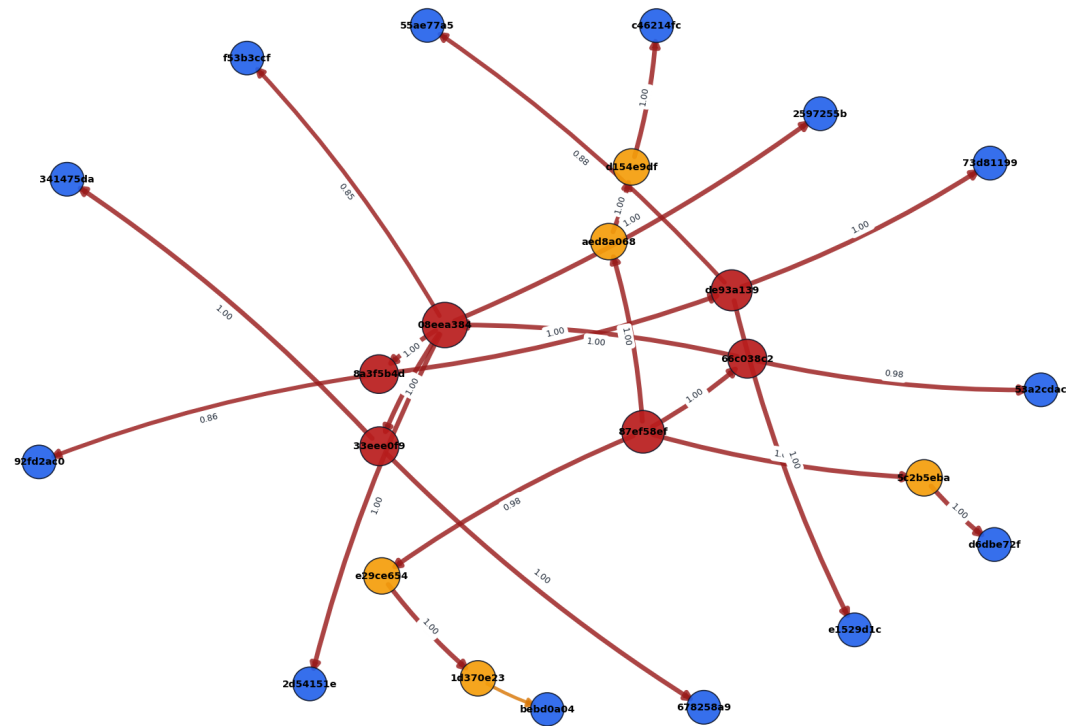


Hist



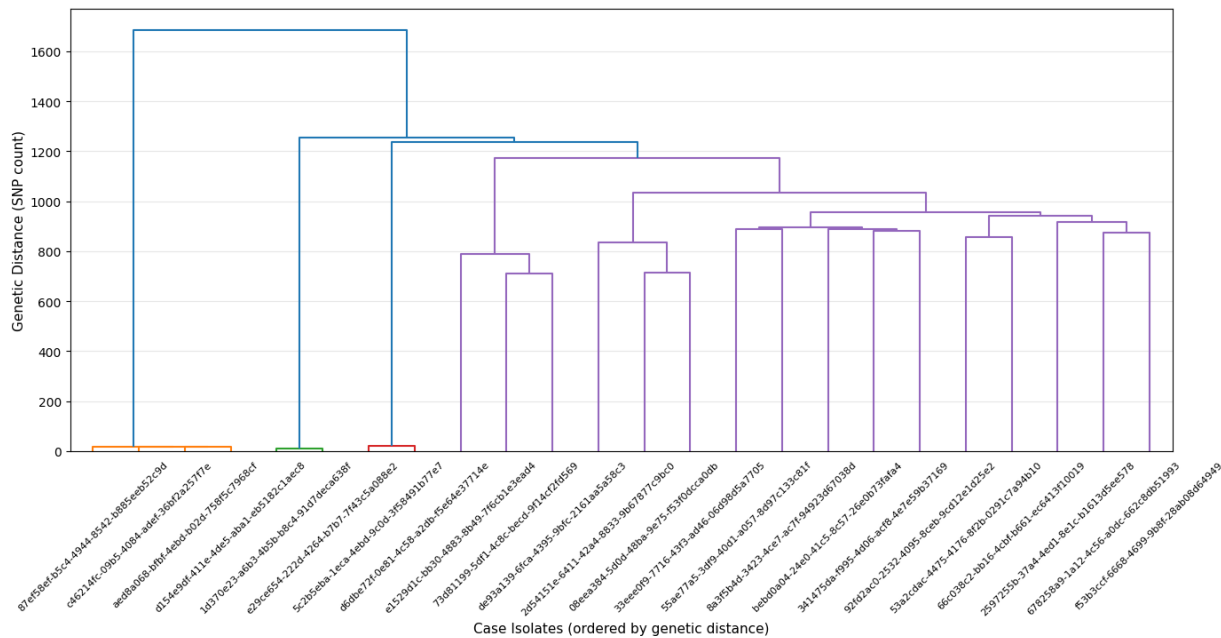
Tree

Transmission Tree (Derived From Posterior Network)

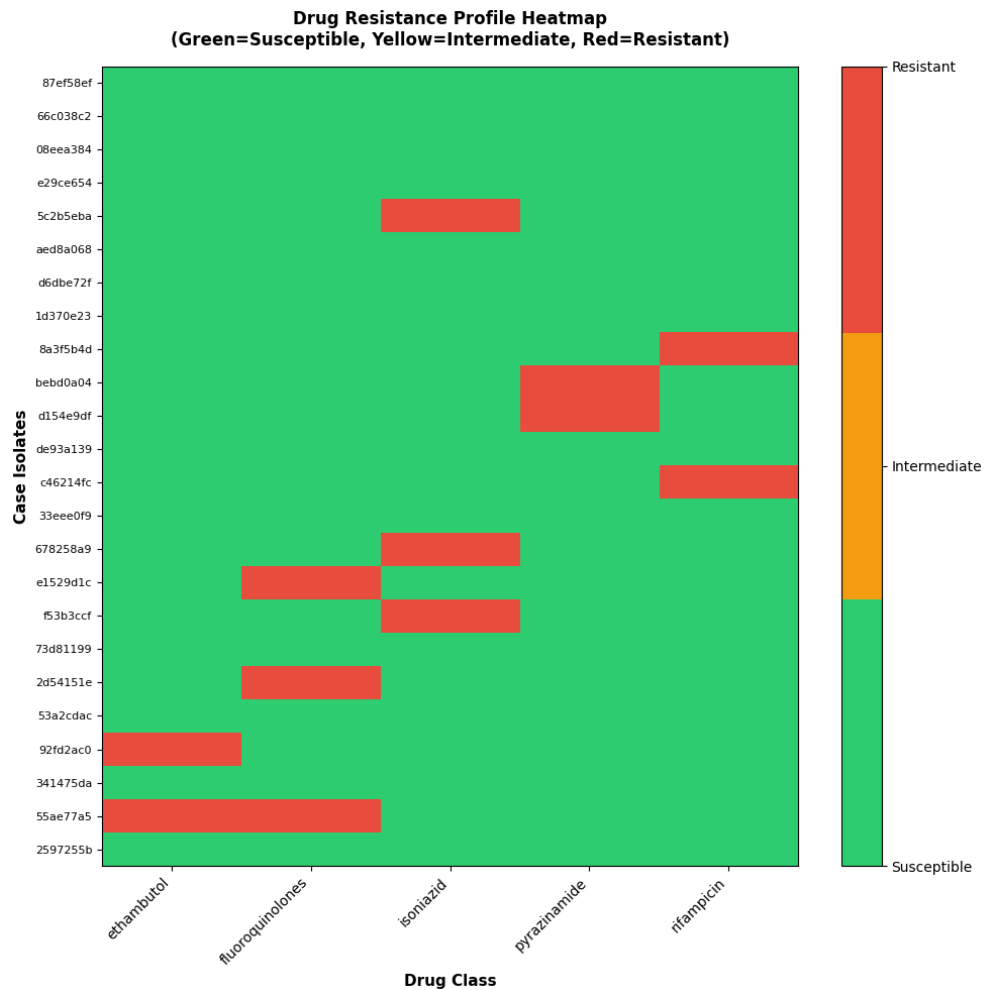


Phylo

Phylogenetic Tree - TB Case Isolates
(Hierarchical Clustering by SNP Distance)



Resistance



Data Provenance

This report combines outbreaker outputs with surveillance KPIs, lineage/DR validation, secondary engine readiness, and cross-method clustering comparison artifacts available at generation time.

Outbreaker summary timestamp: 2026-05-09T12:00:45Z

Transmission network timestamp: 2026-05-09T12:00:45Z

Lineage/DR validation timestamp: 2026-05-09T11:18:32.008979+00:00

Secondary engine validation timestamp: 2026-05-08T22:40:58Z

Method comparison timestamp: 2026-05-09T11:19:28.878816+00:00